



LIFELINE MEDICAL TECHNOLOGIES  
**Precision Analgesia**

# The **Active Safety Net** for Inpatient Opioid Prescribing

A SMART on FHIR Clinical Decision Support engine that intercepts dangerous opioid orders *before* they reach the patient — embedded silently inside Epic and Cerner.



ENTERPRISE B2B SAAS



FDA NON-DEVICE CDS EXEMPT



OPIOID SAFETY

CONFIDENTIAL — FOR INVESTOR REVIEW ONLY



## THE \$95 BILLION BLIND SPOT

## Hospitals are writing opioid orders **without a safety net.**

**\$95B+**

Total U.S. hospital opioid liability exposure — settlements, verdicts, and regulatory penalties

American Hospital Association

**\$303M**

Single "Failure to Monitor" verdict (Asante Rogue Regional). Malpractice is now a nine-figure systemic threat.

Oregon Court of Appeals, 2023

**46%**

Floor patients experience opioid-induced respiratory depression. Manual nursing checks miss the vast majority.

PRODIGY Trial, NEJM

**Alert Fatigue Is the Norm**

Epic's native opioid warnings are dismissed **90–95% of the time**. Clinicians have been trained to click past them. The system meant to protect patients has been optimized away.

**\$651M in Abatement Funds — Unclaimed**

The National Opioid Settlement created a **\$651M acute care hospital fund** specifically for CDS technology. Filing window is open. Hospitals can pay for our integration with settlement money.



## WHY EXISTING TOOLS FAIL

# The difference between **reactive** and **proactive**.

## STATUS QUO

## The Passive Calculator

- ✗ Doctor must consciously *remember* to open MDCalc or a calculator app
- ✗ Manually types in creatinine, weight, opioid doses — 45 seconds of friction
- ✗ Used only 5% of the time. Used in the cafeteria, not at the bedside.
- ✗ Boolean alerts ("MME > 90") dismissed in under 1 second
- ✗ No pharmacogenomics, cross-tolerance, or organ-specific gating

## RESULT

**Optional. Ignored. Liability.**

VS

## PRECISION ANALGESIA

## The Active Safety Net

- ✓ Silently monitors CPOE in the background via CDS Hooks
- ✓ Auto-pulls weight, labs, meds from FHIR — **zero manual entry**
- ✓ Intercepts dangerous orders *at the moment of signing*
- ✓ Returns specific, actionable guidance (not just "warning")
- ✓ CYP2D6 pharmacogenomics, OIRD scoring, renal/hepatic gating

## RESULT

**Mandatory. Specific. Defensible.**



HOW IT WORKS

A three-layer engine embedded in every opioid order.

LAYER 1 — BUILT

**FHIR Ingestion**  
Patient context auto-loads via SMART on FHIR OAuth. Pulls age, weight, creatinine, active meds, allergies, genotype panel. **Zero manual entry.**  

R4 Patient

Observation

MedicationRequest

LAYER 2 — BUILT

**Clinical Pipeline**  
16 deterministic safety rules in <100ms: OME/MME, OIRD risk tier, renal/hepatic gating, pharmacogenomic dose adjustment, CDC naloxone mandate.  

CDS Hooks

PRODIGY

CDC 2022

LAYER 3 — SCAFFOLDED

**FHIR Write-Back**  
1-click "Draft Order" injects safe alternative into Epic's CPOE cart. "Export to Chart" writes a billable consult note as a DocumentReference.  

MedicationRequest POST

DocumentReference

**The Traceability Ledger — Our FDA Criterion 4 Shield**  
Every recommendation shows the exact formula, conversion factor, and guideline citation. The clinician can independently verify the math. This is legally defensible — not a black box.

FDA NON-DEVICE CDS



## CLINICAL MODULES

## Six tools. One workflow. Zero new logins.



### Risk Assessment

OME/MME calculator, OIRD risk tier (PRODIGY-aligned), organ dose gating, naloxone prompt at  $\geq 50$  MME, manual data override form.



### Opioid Tapering

CDC 2022 taper schedule generator. 10%/2-week protocol. One-click copy to EHR note.



### OUD Consult

BUPE/methadone/naltrexone protocols. COWS, DSM-5, aberrant behavior screening. Auto-structured consult note.



### Patch Conversion

Transdermal fentanyl/buprenorphine calculators with organ-adjusted dosing. Prevents the most common ICU conversion error.



### Screening Suite

DAST-10, ASSIST, ORT with sex-adjusted scoring. Reads FHIR nursing flowsheets before prompting manual entry.



### IV Infusion

PCA 1-hr/4-hr limit calculator. Continuous drip converter with renal-adjusted dosing. Prevents common ICU programming errors.



### Manual Override Form — "Dirty Data Defense"

Clinicians correct/supplement FHIR data in real-time. 6 sections, EHR/Manual/Missing badges.



### Copy-to-Note Workflow

All outputs copy as structured clinical text. Paste into Epic note. Citrix/VDI compatible.



## REGULATORY &amp; SECURITY MOATS

## We bypassed the two graveyards **before we started building.**



### THE FDA MOAT Non-Device CDS Exemption

FDA's January 2026 guidance created a "Non-Device CDS" exemption for deterministic rules a clinician can independently verify. Our Traceability Ledger satisfies **Criterion 4** — engineered from day one.

**Saves ~\$2M + 2 Years**

vs. 510(k) SaMD clearance required for AI/ML competitors



### THE CISO MOAT Zero-PHI Architecture

All calculations happen in-memory, discarded after the HTTP response. **No database. No PHI at rest. No HIPAA BAA complexity.** Every hospital CISO's nightmare — a data breach — is architecturally impossible.

**Ships past security review in weeks, not years**



### THE INTEGRATION MOAT SMART on FHIR + CDS Hooks

We embed natively via Epic's published APIs. We never ask hospitals to change their workflows. Once a validated clinical system is integrated, hospital IT will never unwind it.



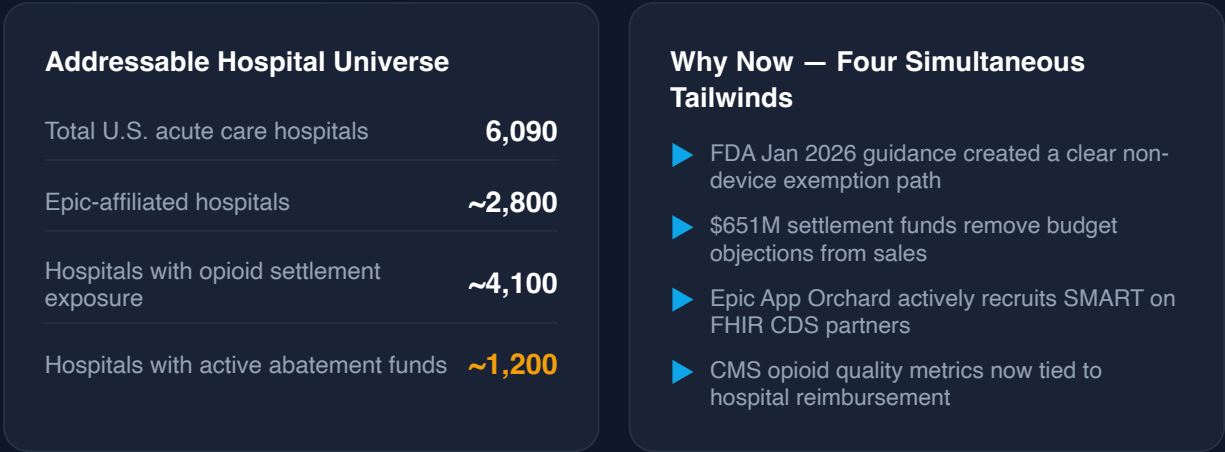
### THE FUNDING MOAT Settlement-Funded Sales

The **\$651M National Opioid Settlement** is earmarked for CDS technology. Hospitals pay with settlement money — removing the capital budget approval from the sales cycle entirely.



TOTAL ADDRESSABLE MARKET

A massive, underserved acute care opportunity.





COMPETITIVE LANDSCAPE

We thread the needle between every incumbent.

| COMPETITOR                                        | APPROACH                                                                        | FATAL WEAKNESS                                           | OUR WEDGE                                                            |
|---------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------|
| <b>Epic / FDB Alerts</b><br>Native EHR            | Boolean pop-ups on every order                                                  | 90–95% dismiss rate. Alert fatigue is a clinical crisis. | Multi-factor engine. Specific. Non-dismissible.                      |
| <b>MDCalc / UpToDate</b><br>SMART calculators     | Reactive reference the doctor must launch                                       | 5% usage rate. Used in the cafeteria, not the CPOE.      | Proactive. Monitors CPOE silently. Fires before the order is signed. |
| <b>Bamboo / NarxCare</b><br>PDMP database         | Narx Score from retail pharmacy fills                                           | Outpatient only. Useless for ICU. Black-box backlash.    | Owns inpatient. Transparent math. Works where NarxCare doesn't.      |
| <b>AI/ML CDS Startups</b><br>Predictive AI        | ML overdose risk models, hundreds of EHR variables                              | FDA 2026 requires 510(k). \$2M+ / 2 years. CISOs refuse. | Deterministic. Non-SaMD. CISO-safe. Pre-cleared.                     |
| <b>Precision Analgesia ✓</b><br>SMART on FHIR CDS | Proactive CDS Hook engine, FHIR auto-population, deterministic transparent math | —                                                        | Inpatient. Proactive. FDA-exempt. CISO-safe. Settlement-funded.      |





## REVENUE

From an **iOS beta** to a **\$250K/year hospital contract**.

## PRIMARY REVENUE

### Enterprise SaaS — Health System Licensing

# \$150K–\$350K

Annual contract per hospital system — priced as a fraction of one prevented malpractice verdict.

|                             |                   |
|-----------------------------|-------------------|
| Implementation & onboarding | \$15–30K one-time |
| Base SaaS license           | \$120–250K/yr     |
| Pharmacogenomics tier       | +\$50K/yr         |
| Settlement documentation    | +\$25K/yr         |

## SECONDARY REVENUE

### iOS App — Individual Clinicians

Currently free (TestFlight beta). Serves as top-of-funnel — a hospital's CMO downloads the app, uses it at home, champions the enterprise integration. Freemium → institutional subscription path.

### Unit Economics (Per Hospital System)

Avg. single preventable opioid verdict **\$8–15M**

Annual cost of our integration **\$150–350K**

Hospital ROI (1 prevented event / 10 yrs) **~30x**

Settlement funds available to pay us **\$651M**

### Go-To-Market Motion

- 1 **Epic App Orchard** → inbound discovery from 2,800 hospital network
- 2 **CMIO champion** path — iOS app creates awareness, CMIO requests enterprise eval
- 3 **Settlement framing** — hospital legal identifies abatement funding, kills budget objection
- 4 **Shadow Mode** — 90-day silent run proves error detection rate before go-live



## WHERE WE ARE TODAY

## Engine built. Ready for the track.

**22,991**

Lines of production  
code (frontend +  
backend)

**25**

Clinical React  
components (6 major  
workflow modules)

**14/14**

Backend clinical logic  
tests passing (pytest)

**16**

Clinical API endpoints  
(rate limited + JWT  
auth)

**✓ Built & Verified**

- ✓ Full SMART on FHIR R4 patient context ingestion
- ✓ 16-rule Safety Engine: OME, OIRD, renal/hepatic, pharmacogenomics
- ✓ WCAG 2.1 A accessibility (screen readers, keyboard nav, VDI-compliant)
- ✓ Rate limiting + JWT token validation on all 16 clinical routes
- ✓ FHIR Write-Back scaffolding (MedicationRequest + DocumentReference)
- ✓ iOS app in TestFlight beta with real clinical users

**🚀 Next 30 Days (Post-Seed)**

- Generate JWKS keys → register at **fhir.epic.com**
- Prove FHIR R4 on Epic Sandbox synthetic patients
- Wire FHIR Write-Back → live 1-click Naloxone co-prescribing
- AKI trending — last 3 creatinine values, not just today's
- Submit to Epic App Orchard — activate inbound discovery
- Sign first pilot hospital for Shadow Mode deployment



FUNDRAISE

Seed Round — putting tires on the Ferrari.

RAISING

\$2.5M

Seed Round — 18-month runway to first signed hospital contracts

Use of Funds

|                                          |              |
|------------------------------------------|--------------|
| Epic Sandbox → Production Integration    | \$600K (24%) |
| Clinical Validation & Pilot Hospital     | \$700K (28%) |
| Engineering (AKI, CDS Hooks, Write-Back) | \$750K (30%) |
| Enterprise Sales & CMIO Outreach         | \$450K (18%) |

18-Month Milestones

- MONTH 3  
Epic Sandbox registration, JWKS keys, FHIR R4 synthetic test passing
- MONTH 6  
First pilot hospital signed. Shadow Mode live. CDS Hooks on medication-prescribe.
- MONTH 9  
Shadow Mode data proves error detection rate. First \$250K contract signed. App Orchard listed.
- MONTH 18  
5+ hospital contracts. Series A raise. SOC 2 Type II audit.

The Settlement-Funded Sales Insight

With the \$651M hospital abatement fund, hospitals pay for our integration **without a capital budget approval**. This compresses an 18-month enterprise sales cycle to 90 days. Our seed capital funds us to the first 5 settlement-funded contracts.

## THE VERDICT

**You are not buying a calculator.  
You are buying a hospital safety  
mandate.**

Every hospital without this tool is writing opioid orders without a safety net. One prevented verdict pays for 50 years of our contract. The clinical engine is built. The regulatory path is clear. The settlement funds are ready.

WEBSITE

[opioidprecision.com](https://opioidprecision.com)

IOS BETA

[TestFlight](#)

COMPANY

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